

Commutative Algebra

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Abstract

Math 614

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Chapter 1

Introduction

Lecture 1: Motivations and Affine Varieties

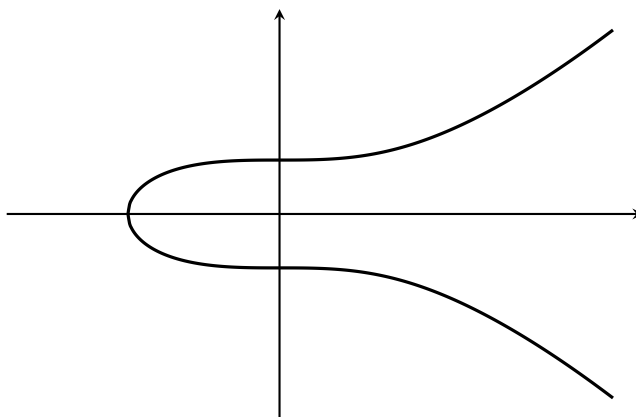
There are many motivations for studying commutative algebra. Here are two of the most important ones: 1 Sep. 9:00

1.1 Algebraic Geometry

Definition 1.1.1 (Affine algebraic variety). Let $f_1, \dots, f_r \in \mathbb{C}[x_1, \dots, x_n]$. An affine algebraic variety over $\mathbb{C}$ is the common zero locus of $f_1, \dots, f_r$. i.e.

$$X = \{p = (a_1, \dots, a_n) \in \mathbb{C}^n \mid f_1(p) = \dots = f_r(p) = 0\}$$

$$X = \{(x, y) \in \mathbb{R}^2 : y^2 = x^3 + 1\}.$$



Definition 1.1.2. For each affine algebraic variety $X \subseteq \mathbb{C}^n$, we can define the ideal of X as

$$I(X) := \{f \in \mathbb{C}[x_1, \dots, x_n] \mid f(p) = 0, \forall p \in X\}$$

Definition 1.1.3.

$$\mathcal{O}(X) := \mathbb{C}[x_1, \dots, x_n]/I(X)$$

is called the *coordinate ring* of X .

Remark. Here is another way to think of $\mathcal{O}(X)$. Consider the following diagram:

$$\begin{array}{ccc}
 \mathbb{C}[x_1, \dots, x_n] & \xrightarrow{f \mapsto (p \mapsto f(p))} & \{ \text{functions } X \rightarrow \mathbb{C} \} \\
 \downarrow & \searrow & \uparrow \\
 \mathcal{O}(X) & \xrightarrow{\sim} & \{ \text{polynomial functions on } X \}
 \end{array}$$

Then $\mathcal{O}(X)$ can be identified with polynomial functions on X .

Theorem 1.1.1 (Nullstellensatz).

1. There is a bijection

$$\begin{aligned}
 X &\xleftrightarrow{\sim} \{ \mathfrak{m} \subset \mathcal{O}(X) \mid \mathfrak{m} \text{ maximal} \}, \\
 p = (a_1, \dots, a_n) \in X &\longleftrightarrow \mathfrak{m}_p = (\bar{x}_1 - a_1, \dots, \bar{x}_n - a_n).
 \end{aligned}$$

2. Let $X = V(f_1, \dots, f_r)$ be an affine algebraic variety. Then $I(X) = \{ f \in \mathbb{C}[x_1, \dots, x_n] \mid f^m \in (f_1, \dots, f_r) \text{ for some } m \in \mathbb{Z}_{>0} \}$.

Remark. $\mathfrak{m}_p$ is a maximal ideal since

$$\begin{aligned}
 \mathfrak{m}_p &= \ker(\text{ev}_p : \mathcal{O}(X) \rightarrow \mathbb{C}) \\
 &= \{ f \in \mathcal{O}(X) \mid f(p) = 0 \}
 \end{aligned}$$

1.2 Number theory

The basic problem in number theory is to determine the set of solutions

$$\{ p = (a_1, \dots, a_n) \in \mathbb{Z}^n \mid f_1(p) = \dots = f_r(p) = 0 \}$$

Example (Fermat Last Theorem). Let $n \geq 3$. Then the equation

$$x^n + y^n = z^n$$

has no nontrivial integer solutions.

An elementary approach to this problem is to factorize $z^n - x^n$:

$$y^n = z^n - x^n = (z - x)(z - \zeta x) \cdots (z - \zeta^{n-1} x)$$

where $\zeta = e^{2\pi i/n}$ is a primitive n -th root of unity.

Lamé (1847) “Proved” Fermat’s Last Theorem by comparing factorizations on LHS and RHS, using erroneous assumption that $\mathbb{Z}[\zeta]$ is a UFD.

Kummer Introduced the notion of an ideal to resolve a version of unique factorization and salvage part of Lamé’s proof.

Theorem 1.2.1. For any ideal $0 \neq I \subseteq \mathbb{Z}[\zeta]$, there exists a factorization, unique up to reordering factors, of the form

$$I = \mathfrak{p}_1 \mathfrak{p}_2 \cdots \mathfrak{p}_r$$

where $\mathfrak{p}_i \subseteq \mathbb{Z}[\zeta]$ are prime ideals.

Remark. This is true more generally for rings of integers $\mathcal{O}_K \subseteq K$ of any finite extension $K/\mathbb{Q}$, where

$$\mathcal{O}_K = \{a \in K \mid \text{there is a monic polynomial } f(x) \in \mathbb{Z}[x] \text{ such that } f(a) = 0\}$$

Commutative algebra gives a common framework for studying algebraic geometry, number theory, etc. For example, the embedding $\mathbb{Z} \hookrightarrow \mathcal{O}_K$ has finite branch cover, which helps us to understand number theory from the perspective of commutative algebra.

1.3 Course Conventions

- Rings R are commutative with identity $1 \neq 0$. For the zero ring, $1 = 0$ is allowed.
- Ring homomorphisms sends $1 \mapsto 1$.
- A *subring* of R is a subset $S \subseteq R$ such that $1 \in S$ and S is closed under addition, multiplication, and additive inverses.

Here are some reminders:

Definition 1.3.1. An ideal $\mathfrak{p} \subset R$ is *prime* if

1. $\mathfrak{p} \neq R$.
2. If $xy \in \mathfrak{p}$, then $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$.

Definition 1.3.2. An ideal $\mathfrak{m} \subset R$ is *maximal* if

1. $\mathfrak{m} \neq R$.
2. There does not exist any ideal $I \subseteq R$ with $\mathfrak{m} \subsetneq I \subsetneq R$.

Lemma 1.3.1.

1. An ideal $\mathfrak{p} \subset R$ is prime if and only if $R/\mathfrak{p}$ is an integral domain.
2. An ideal $\mathfrak{m} \subset R$ is maximal if and only if $R/\mathfrak{m}$ is a field.
3. Every maximal ideal is prime.
4. For any proper ideal $I \subsetneq R$, there exists a maximal ideal $\mathfrak{m} \subset R$ such that $I \subseteq \mathfrak{m}$.

The motivating geometric idea is to think of a ring R as the “ring of algebraic functions” on a suitable space, analogous to $X \leftrightarrow \mathcal{O}(X)$.

Nullstellensatz suggests studying maximal ideals

$$\text{mSpec}(R) = \{\mathfrak{m} \subset R \mid \mathfrak{m} \text{ is maximal}\}$$

But the problem is that $\text{mSpec}(R)$ does not encode morphisms between objects of interest. We first define morphisms between varieties.

Definition 1.3.3. A *morphism* $\phi : X \rightarrow Y$ of affine algebraic varieties, where $X \subseteq \mathbb{C}^n$ and $Y \subseteq \mathbb{C}^m$,

is a map of the form

$$\varphi = (f_1, \dots, f_m) : X \rightarrow Y, \quad f_i \in \mathbb{C}[x_1, \dots, x_n]$$

Example. Consider the previous example, we have the following:

$$\begin{array}{ccc} X = \{(x, y) \in \mathbb{C}^2 \mid y = x^2\} & & \psi(t) = (t, t^2) \\ \varphi \downarrow \uparrow \psi & & \\ Y = \mathbb{C} & & \varphi(x, y) = x \end{array}$$

It is easy to see that φ and ψ are inverse morphisms. In this case X and Y are isomorphic.

Remark.

1. It is easy to check that morphisms of varieties have expected properties, i.e. composition, associative, identity. We then get a category $\text{Aff}_{\mathbb{C}}$ of affine varieties over $\mathbb{C}$.
2. We use the notation $\text{Hom}_{\text{Aff}_{\mathbb{C}}}(X, Y) = \{\text{morphisms } X \rightarrow Y\}$. Then

$$\mathcal{O}(X) \cong \{\text{Polynomial functions on } X\} = \text{Hom}_{\text{Aff}_{\mathbb{C}}}(X, \mathbb{C})$$

Definition 1.3.4. Let $\varphi : X \rightarrow Y$ be a morphism of affine varieties. Then φ induces a map

$$\begin{aligned} \varphi^* : \mathcal{O}(Y) &\longrightarrow \mathcal{O}(X), \\ f : Y &\rightarrow \mathbb{C} \longmapsto f \circ \varphi : X \rightarrow \mathbb{C}. \end{aligned}$$

Moreover, this is easy to see that this is a $\mathbb{C}$ -algebra homomorphism.

Example. Return to the previous example, we have

$$\begin{aligned} \varphi^* : \mathcal{O}(Y) = \mathbb{C}[t] &\longrightarrow \mathcal{O}(X) = \frac{\mathbb{C}[x, y]}{(y - x^2)}, \\ t &\longmapsto \bar{x}. \\ \psi^* : \frac{\mathbb{C}[x, y]}{(y - x^2)} &\longrightarrow \mathbb{C}[t], \\ \bar{x} &\longmapsto t, \\ \bar{y} &\longmapsto t^2. \end{aligned}$$

Lemma 1.3.2. The map $(-)^* : \text{Hom}_{\text{Aff}_{\mathbb{C}}}(X, Y) \rightarrow \text{Hom}_{\mathbb{C}\text{-alg}}(\mathcal{O}(Y), \mathcal{O}(X))$ is a bijection.

Proof Sketch. Construct an inverse to $(-)^*$ as follows. Given $f : \mathcal{O}(Y) \rightarrow \mathcal{O}(X)$, i.e. $f : \mathbb{C}[y_1, \dots, y_m]/I(Y) \rightarrow \mathbb{C}[x_1, \dots, x_n]/I(X)$, we know that it is determined by the images of $\bar{y}_i$. Let the image of $\bar{y}_i$ be $\bar{\varphi}_i$ for some $\varphi_i \in \mathbb{C}[x_1, \dots, x_n]$.

(HW 1): Check: $(\varphi = (\varphi_1, \dots, \varphi_m) : X \rightarrow Y) \in \text{Hom}_{\text{Aff}_{\mathbb{C}}}(X, Y)$, and this construction is inverse to $(-)^*$. ■

Lemma 1.3.3. Let $\varphi : X \rightarrow Y$ be a morphism of affine varieties, where $X \subseteq \mathbb{C}^n$ and $Y \subseteq \mathbb{C}^m$. Under the bijections from Nullstellensatz, we have

$$X \cong \text{mSpec}(\mathcal{O}(X)) \quad \text{and} \quad Y \cong \text{mSpec}(\mathcal{O}(Y)),$$

then φ corresponds to the map

$$\mathrm{mSpec}(\mathcal{O}(X)) \rightarrow \mathrm{mSpec}(\mathcal{O}(Y)), \quad \mathfrak{m} \mapsto (\varphi^*)^{-1}(\mathfrak{m})$$

Proof. Let $p = (a_1, \dots, a_n) \in X$, then $\mathfrak{m}_p = (\bar{x}_1 - a_1, \dots, \bar{x}_n - a_n) \in \mathrm{mSpec}(\mathcal{O}(X))$. Then

$$\begin{aligned} (\varphi^*)^{-1}(\mathfrak{m}_p) &= \{g \in \mathcal{O}(Y) \mid \varphi^*(g) \in \mathfrak{m}_p\} \\ &= \{g \in \mathcal{O}(Y) \mid g(\varphi(p)) = 0\} \\ &= \mathfrak{m}_{\varphi(p)} \end{aligned}$$

■

Lecture 1: Motivations and Affine Varieties

As previously seen. We have seen last time that there is a bijection

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$$\left\{ \begin{array}{c} \text{morphisms between affine varieties over } \mathbb{C} \\ \phi : X \rightarrow Y \end{array} \right\} \iff \left\{ \begin{array}{c} \mathbb{C}\text{-algebra homomorphisms} \\ \varphi^* : \mathcal{O}(Y) \rightarrow \mathcal{O}(X) \end{array} \right\}.$$

If we identify X with $\mathrm{mSpec}(\mathcal{O}(X))$ and Y with $\mathrm{mSpec}(\mathcal{O}(Y))$, then ϕ corresponds to the map that sends a maximal ideal $\mathfrak{m} \subset \mathcal{O}(X)$ to the maximal ideal $(\varphi^*)^{-1}(\mathfrak{m})$.

Now, back to general rings. For any ring homomorphism $f : R \rightarrow S$, the above idea suggests us to define

$$\begin{aligned} \mathrm{mSpec}(S) &\rightarrow \mathrm{mSpec}(R) \\ \mathfrak{m} &\mapsto f^{-1}(\mathfrak{m}) \end{aligned}$$

The problem is that it is not well-defined. If $\mathfrak{m}$ is a maximal ideal of S , then $f^{-1}(\mathfrak{m})$ is not necessarily a maximal ideal of R . For example, consider the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$. The zero ideal (0) is a maximal ideal of $\mathbb{Q}$, but its preimage under the inclusion is (0) in $\mathbb{Z}$, which is not maximal.

However, if $\mathfrak{q} \subset S$ is a prime ideal, then $f^{-1}(\mathfrak{q})$ is a prime ideal of R . To see this, we have

$$\begin{aligned} xy \in f^{-1}(\mathfrak{q}) &\Rightarrow f(x)f(y) \in \mathfrak{q} \\ &\Rightarrow f(x) \in \mathfrak{q} \text{ or } f(y) \in \mathfrak{q} \\ &\Rightarrow x \in f^{-1}(\mathfrak{q}) \text{ or } y \in f^{-1}(\mathfrak{q}) \end{aligned}$$

Definition 1.3.5. Let R be a ring. The *prime spectrum* of R , denoted by $\mathrm{Spec}(R)$, is the set of all prime ideals of R . i.e.

$$\mathrm{Spec}(R) = \{\mathfrak{p} \subset R : \mathfrak{p} \text{ is a prime ideal}\}$$

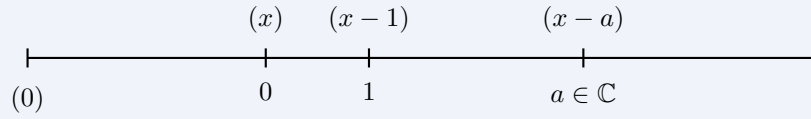
Given ring homomorphism $f : R \rightarrow S$, we have a well-defined map $\mathrm{Spec}(f) : \mathrm{Spec}(S) \rightarrow \mathrm{Spec}(R)$ given by $\mathfrak{q} \mapsto f^{-1}(\mathfrak{q})$.

Example.

- (0) If $R = 0$, then $\mathrm{Spec}(R) = \emptyset$.
- (1) If $R = k$ is a field, then $\mathrm{Spec}(R) = \{(0)\}$.
- (2) If k is a field, then $\mathrm{Spec}(k[x]/(x^2)) = \{(\bar{x})\}$.
- (3) If R is a PID, then $\mathrm{Spec}(R) = \{(0)\} \cup \{(p) \mid p \in R \text{ is irreducible}\}$.

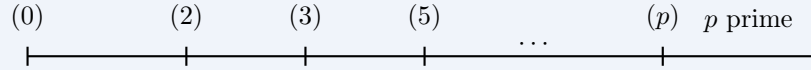
Example. The geometry of $\mathrm{Spec}(\mathbb{C}[x])$ look like

$\text{Spec}(\mathbb{C}[x])$



and the geometry of $\text{Spec}(\mathbb{Z})$ looks like

$\text{Spec}(\mathbb{Z})$

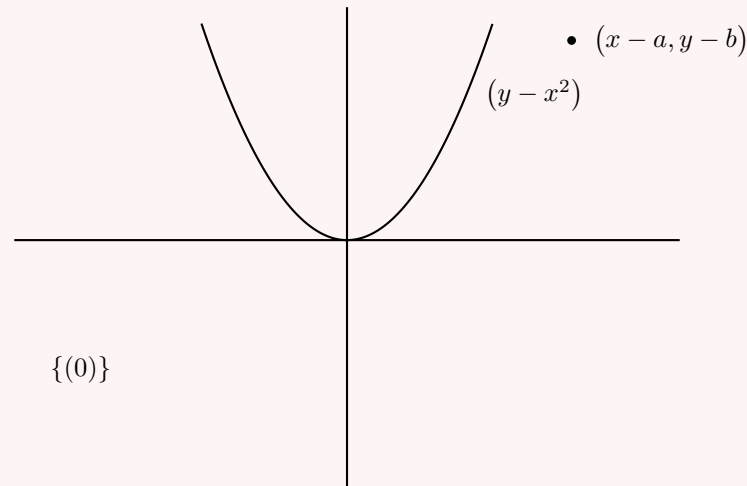


This (0) is called the *generic point* of $\text{Spec}(\mathbb{C}[x])$ or $\text{Spec}(\mathbb{Z})$. We have $\overline{(0)} = \text{Spec}(\mathbb{C}[x])$ or $\text{Spec}(\mathbb{Z})$ in the later topology we define.

Problem 1.3.1 (HW 1). Prove that

$$\text{Spec}(\mathbb{C}[x, y]) = \{(x - a, y - b) \mid a, b \in \mathbb{C}\} \cup \{f \mid f \in \mathbb{C}[x, y] \mid f \text{ is irreducible}\} \cup \{(0)\}$$

Picture:



1.4 Functions on $\text{Spec}(R)$

Motivation: Let $X \subset \mathbb{C}^n$ be an affine variety. Its coordinate ring is $\mathcal{O}(X) = \mathbb{C}[x_1, \dots, x_n]/I(X) = \text{Hom}_{\text{Aff}_\mathbb{C}}(X, \mathbb{C})$. Under the bijection $X \cong \text{mSpec}(\mathcal{O}(X))$, given $f \in \text{Hom}_{\text{Aff}_\mathbb{C}}(X, \mathbb{C})$ and $p \in X$, we have $f(p) \in \mathbb{C}$. On the other hand, given $\mathfrak{m} \in \text{mSpec}(\mathcal{O}(X))$, its corresponding value should be $f \bmod \mathfrak{p} \in \mathcal{O}(X)/\mathfrak{m} \cong \mathbb{C}$, which corresponds to the same value to $f(p)$ through the isomorphism. This motivates the following definition:

Definition 1.4.1. Let R be a ring. Let $\mathfrak{p} \in \text{Spec}(R)$. The *residue field* of R at $\mathfrak{p}$ is defined to be the field

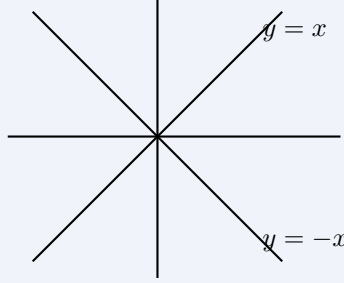
$$\kappa(\mathfrak{p}) := \text{Frac}(R/\mathfrak{p})$$

Given $f \in R$, the *value* of f at $\mathfrak{p}$ is

$$f(\mathfrak{p}) := f \bmod \mathfrak{p} \in R/\mathfrak{p}$$

Remark. When $R = \mathcal{O}(X)$, and $\mathfrak{p}$ is a maximal ideal, then $f(\mathfrak{p}) \in \mathbb{C}$ is the usual value of f at the point $p \in X$ corresponding to the maximal ideal $\mathfrak{p}$.

Example. Let $R = \mathbb{C}[x, y]/(x^2 - y^2)$.



Then $\text{Spec}(R)$ consists of the following prime ideals:

1. $\mathfrak{p} = (\overline{x - a}, \overline{y - a}) \in \text{mSpec}(R)$.
2. $\mathfrak{p} = (\overline{x + a}, \overline{y - a}) \in \text{mSpec}(R)$.
3. $\mathfrak{p} = (\overline{x - y}) \in \text{mSpec}(R)$.
4. $\mathfrak{p} = (\overline{x + y}) \in \text{mSpec}(R)$.

We consider the first case by taking $\mathfrak{p} = (\overline{x}, \overline{y})$. For $f \in R$, the value at $\mathfrak{p}$ is

$$f(\mathfrak{p}) = f \mod (\overline{x}, \overline{y}) \in R/(\overline{x}, \overline{y}) \cong \mathbb{C}$$

In the third case, we have

$$f(\mathfrak{p}) = f \mod (\overline{x - y}) \in R/(\overline{x - y}) \cong \mathbb{C}[x, y]/(x - y) \cong \mathbb{C}[x]$$

For instance, if $f = 5 + \overline{xy}$, then $f(\mathfrak{p}) = 5 + \overline{xy} \mod (\overline{x - y}) \equiv 5 + \overline{x^2} \in \mathbb{C}(x) = \kappa(\mathfrak{p})$.

If we take $R = \mathbb{Z}$, and take $f \in R$, then the prime ideals of $\mathbb{Z}$ consists of:

1. $(p) \in \text{mSpec}(R)$, $p \in \mathbb{Z}$ prime. Then $f(\mathfrak{p}) = f \mod (p) \in \mathbb{Z}/(p) \cong \mathbb{F}_p$.
2. $(0) \in \text{Spec}(R)$. Then $f(\mathfrak{p}) = f \mod (0) \in \text{Frac } \mathbb{Z} = \mathbb{Q}$.

Example (Warning). A nonzero function can vanish at all points of $\text{Spec}(R)$. Consider $R = k[x]/(x^2)$, then $\text{Spec}(R) = \{(\overline{x})\}$. For $f = \overline{x} \in R$, we have $f(\mathfrak{p}) = f \mod (\overline{x}) = 0 \in R/(\overline{x}) \cong k$.

But note that $f^2 = 0$. (Later we'll see this is a general phenomenon: if $f(\mathfrak{p}) = 0$ for all $\mathfrak{p} \in \text{Spec}(R)$, then f is nilpotent.)

1.5 Topology on $\text{Spec}(R)$

The idea to define a topology on $\text{Spec}(R)$ is that the locus where functions vanish “ought” to be closed sets.

Definition 1.5.1. Let R be a ring, and let $T \subset R$ be a subset. Then the *vanishing locus* of T is defined to be

$$\begin{aligned} V(T) &:= \{\mathfrak{p} \in \text{Spec}(R) \mid f(\mathfrak{p}) = 0 \text{ for all } f \in T\} \\ &= \{\mathfrak{p} \in \text{Spec}(R) \mid T \subset \mathfrak{p}\} \end{aligned}$$

There are some basic properties of $V(T)$:

Proposition 1.5.1.

- (1) Let (T) be the ideal generated by T . Then $V(T) = V((T))$.
1. If subsets $T \subset T'$, then $V(T') \subset V(T)$.
- (3) $V(0) = \text{Spec}(R)$.
- (4) $V(R) = \emptyset$.
- (5) For any family of ideals $\{I_i\}_{i \in J}$, we have

$$\bigcap_{i \in J} V(I_i) = V\left(\bigcup_{i \in J} I_i\right) = V\left(\sum_{i \in J} I_i\right)$$

- (6) Let $I, J \subset R$ be ideals. Then

$$V(I) \cup V(J) = V(I \cap J) = V(IJ)$$

Proof. We only prove (6). The other properties are easy to check. We always have $I \cdot J \subset I \cap J \subset I \cdot J$. Then, by (2), we have

$$V(I) \cup V(J) \subset V(I \cap J) \subset V(I \cdot J)$$

It remains to show that $V(I \cdot J) \subset V(I) \cup V(J)$. Suppose there is $\mathfrak{p} \in V(I \cdot J)$ but $\mathfrak{p} \notin V(I)$, $\mathfrak{p} \notin V(J)$. Then there exists $f \in I$ and $g \in J$ such that $f(\mathfrak{p}) \neq 0$ and $g(\mathfrak{p}) \neq 0$, which is equivalent to say that $f \notin \mathfrak{p}$ and $g \notin \mathfrak{p}$. But then $fg \in I \cdot J \subset \mathfrak{p}$, which contradicts the fact that $\mathfrak{p}$ is prime. Therefore $V(I \cdot J) \subset V(I) \cup V(J)$. ■

Corollary 1.5.1. There is a topology on $\text{Spec}(R)$, called the *Zariski topology*, whose closed sets are of the form $V(I)$ for some subset $T \subset R$ (or ideals, which are equivalent as in (1) in the previous proposition).

Proof. We have shown that:

- $\emptyset$ and $\text{Spec}(R)$ are closed.
- Closed sets are closed under arbitrary intersections.
- Closed sets are closed under finite unions.

So it is indeed a topology. ■

Remark. There is a natural analogue of the Zariski topology for any variety $X \subset \mathbb{C}^n$, with closed subsets given by subvarieties of X , where

$$X \cong \text{mSpec}(\mathcal{O}(X)) \hookrightarrow \text{Spec}(\mathcal{O}(X))$$

This topology corresponds to the induced topology on $\text{Spec}(\mathcal{O}(X))$. So see this, we have $V(I) \cap \text{mSpec}(\mathcal{O}(X)) = \{\mathfrak{m}_p \mid I \subseteq \mathfrak{m}_p\}$. This is the same as $V_X(I)$, the vanishing locus of I in X .

Definition 1.5.2. For $f \in R$, we have the *distinguished open set* $D(f) \subset \text{Spec}(R)$ defined by

$$\begin{aligned} D(f) &:= \{\mathfrak{p} \in \text{Spec}(R) \mid f(\mathfrak{p}) \neq 0\} \\ &= \{\mathfrak{p} \in \text{Spec}(R) \mid f \notin \mathfrak{p}\} \\ &= \text{Spec}(R) \setminus V(f) \end{aligned}$$

which is open.

Remark. These distinguished open sets $D(f)$ form a basis for the Zariski topology on $\text{Spec}(R)$. If $U \subseteq \text{Spec}(R)$ is open. Write $U = \text{Spec}(R) \setminus V(I)$ for some ideal $I \subset R$. Then we have $U = \bigcup_{f \in I} D(f)$.

Example. Consider $\text{Spec}(\mathbb{C}[x])$. Then the closed sets consist of $\emptyset$, $\text{Spec}(\mathbb{C}[x])$, and $V(f)$ for some nonzero polynomial $f \in \mathbb{C}[x]$ (since $\mathbb{C}[x]$ is a PID). Write $f = c(x - a_1)(x - a_2) \cdots (x - a_n)$, then $V(f) = \{(x - a_1), (x - a_2), \dots, (x - a_n)\}$. So the open sets are $\text{Spec}(\mathbb{C}[x])$, $\emptyset$, and complements of finitely many maximal ideals $(x - a_i)$.



Note. The topology on $\text{Spec}(R)$ is very different (much coarser) than the usual Euclidean topology on $\mathbb{C} \cong \text{mSpec}(\mathbb{C}[x])$. For example, Zariski topology is not Hausdorff (only finitely many $x - a_i$ are excluded). Moreover $(0) \in \text{Spec}(\mathbb{C}[x])$ is not a closed point as what we would expect in the Euclidean topology. In fact, $\overline{(0)} = \text{Spec}(\mathbb{C}[x])$.

1.6 Functoriality

Lemma 1.6.1. Let $f : R \rightarrow S$ be a ring homomorphism. Let $T \subset R$ be a subset. Let $\text{Spec}(f) : \text{Spec}(S) \rightarrow \text{Spec}(R)$. Then $\text{Spec}(f)^{-1}(V(T)) = V(f(T))$.

Proof.

$$\begin{aligned} \mathfrak{p} \in \text{Spec}(f)^{-1}(V(T)) &\Leftrightarrow \text{Spec}(f)(\mathfrak{p}) \in V(T) \\ &\Leftrightarrow f^{-1}(\mathfrak{p}) \in V(T) \\ &\Leftrightarrow T \subset f^{-1}(\mathfrak{p}) \\ &\Leftrightarrow f(T) \subset \mathfrak{p} \\ &\Leftrightarrow \mathfrak{p} \in V(f(T)) \end{aligned}$$

■

Corollary 1.6.1. $\text{Spec}(f) : \text{Spec}(S) \rightarrow \text{Spec}(R)$ is continuous.

Lemma 1.6.2. Let $I \subseteq R$ be an ideal. Let $f : R \rightarrow R/I$ be the quotient map. Then $\text{Spec}(f) : \text{Spec}(R/I) \rightarrow \text{Spec}(R)$ induces a homeomorphism

$$\text{Spec}(R/I) \cong V(I) \subset \text{Spec}(R)$$

Proof. Since primes in R/I are in bijection with primes in R containing I , we have a bijection $\text{Spec}(R/I) \cong V(I)$. It remains to show that this bijection is continuous and its inverse is continuous. We have shown that $\mathfrak{p} \mapsto f^{-1}(\mathfrak{p})$ is continuous. Now we show that this is a closed map. Let $J \subset R/I$ be an ideal. Note that for $\mathfrak{p} \in \text{Spec}(R/I)$, $J \subset \mathfrak{p}$ if and only if $f^{-1}(J) \subset f^{-1}(\mathfrak{p})$. Therefore,

$$\text{Spec}(f)(V(J)) = V(f^{-1}(J)) \cap V(I)$$

which is closed. So $\text{Spec}(f)$ is a closed map. Therefore, $\text{Spec}(f) : \text{Spec}(R/I) \rightarrow V(I)$ is a homeomorphism. ■

Appendix

Appendix A

Finite and integral homomorphisms

A.1 Definitions

Let $\varphi : R \rightarrow S$ be a ring homomorphism. One say that φ is of *finite type* if S is a finitely generated R -algebra. One say that φ is *finite* if S is a finitely generated R -module. One say that φ is *integral* if every element of S is integral over R , i.e., for every $s \in S$, there exists a monic polynomial $f(x) \in R[x]$ such that $f(s) = 0$.

Remark. It is clear that if φ is finite, then it is of finite type. The converse is not true. For example, take the inclusion $R \hookrightarrow R[x]$. Then $R[x]$ is a finitely generated R -algebra, but it is not a finitely generated R -module, for finitely many polynomials only generate bounded degree.

Remark. If φ is of finite type and integral, then it is finite. Let $y_1, \dots, y_r$ generates S as an R -algebra. Since φ is integral, we can write

$$y_i^{d_i} + a_{i,1}y_i^{d_i-1} + \dots + a_{i,d_i} = 0$$

Then it is easy to see that

$$\{y_1^{a_1} \dots y_r^{a_r} \mid 0 \leq a_i \leq d_i - 1\}$$

generates S as an R -module.

Proposition A.1.1. If φ is finite, then it is integral.

Proof. Suppose that $b_1, \dots, b_n$ generate S as an R -module, then for every $y \in S$, write, for each $1 \leq i \leq n$,

$$yb_i = \sum_{j=1}^n a_{i,j} b_j \quad \text{for some } a_{i,j} \in R$$

Let A be the matrix $(a_{i,j})_{1 \leq i,j \leq n}$, then we have

$$(yI - A) \cdot \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0$$

Multiplying by the adjoint matrix of $(yI - A)$, we see that if $D = \det(yI - A)$, then $Db_i = 0$ for every $1 \leq i \leq n$. This implies $D \cdot 1_S = 0$. However, it is clear that we can write

$$D = y^n + c_1 y^{n-1} + \dots + c_n \quad \text{for some } c_i \in R$$

We see that y is integral over R . ■

Remark. It follows from [Proposition A.1.1](#) and the previous remark that if φ is of finite type, then it is finite if and only if it is integral.

A.2 Easy properties

Proposition A.2.1. Let $\varphi : R \rightarrow S$ be an injective, integral homomorphism of integral domains, then R is a field if and only if S is a field.

Proof. Suppose that R is a field, and let $u \in S \setminus \{0\}$. Since φ is integral, we can write

$$u^n + a_1 u^{n-1} + \dots + a_n = 0 \quad \text{for some } a_i \in R$$

We see that $u(u^{n-1} + a_1 u^{n-2} + \dots + a_{n-1}) = -a_n$. Since $u \neq 0$, $a_n \neq 0$. Since R is a field, a_n is invertible in R , we see that $uv = 1$, where

$$v = (-a_n)^{-1}(u^{n-1} + a_1 u^{n-2} + \dots + a_{n-1}) \in S$$

Since it holds for all $u \in S \setminus \{0\}$, we see that S is a field. Conversely, suppose that S is a field, and let $a \in R \setminus \{0\}$. Let $b = a^{-1} \in S$, we want to show that $b \in R$. Since φ is integral, we can write

$$b^n + \alpha_1 b^{n-1} + \dots + \alpha_n = 0 \quad \text{for some } \alpha_i \in R$$

but then $b = -\alpha_1 - \alpha_2 a - \dots - \alpha_n a^{n-1} \in R$. Since it holds for all $a \in R \setminus \{0\}$, we see that R is a field. ■

Proposition A.2.2. Given a ring homomorphism $\varphi : R \rightarrow S$, the subset

$$S' := \{y \in S \mid y \text{ is integral over } R\}$$

is a subring of S . This is the *integral closure* of R in S .

Proof. It is easy to see that $1_s \in S'$. It remains to show that if $y_1, y_2 \in S'$, then $y_1 - y_2, y_1 y_2 \in S'$. Since y_1, y_2 are integral over R , we know that the ring $R[y_1, y_2]$ is finite over R , hence integral over R by [Proposition A.1.1](#). This implies that $y_1 - y_2, y_1 y_2 \in S'$. ■

Proposition A.2.3. Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms. If φ and ψ are of finite type (respectively finite, integral), then so is $\psi \circ \varphi$.

Proof. The assertion is straightforward for finite and finite type morphisms. Suppose now that both φ and ψ are integral. Let $u \in T$, then u is integral over S , so we can write

$$u^n + b_1 u^{n-1} + \dots + b_n = 0 \quad \text{for some } b_i \in S$$

Since $b_1, \dots, b_n$ are integral over R , the ring $R' := R[b_1, \dots, b_n]$ is finite over R . Moreover, $R'[u]$ is finite over R' since u is integral over R' . Hence $R'[u]$ is finite over R , hence integral over R by [Proposition A.1.1](#). ■